

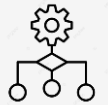
XIAMEN
UNIVERSITY

COM102 – Programming Language C

Chapter 3 - Control Flow (iii) - Lab

Edited By: Dr. Najla'a A. M. Draib

Chapter 3 Control Flow



- 3.1 Branching and Jumps
  - 3.1.1 if statements
  - 3.1.2 switch...case Statement
- 3.2 Loop
  - 3.2.1 while Loop
  - 3.2.2 do .. while Loop
  - 3.2.3 for Loop
- 3.3 Nested Loops

Chapter 3

3.2 Loops

Exercise 1

Read integer sensor readings one by one.

- **Stop** normally when the user enters -1 (sentinel).
- **Skip** (don't count) any odd reading using continue.
- If any reading is greater than 1000, treat it as a **spike** → print a warning and break.
- Finally, **print** how many **even** readings were processed and their average (2 decimal places). **Handle** the case of **zero valid readings**.

Exercise 2

Print a right-triangular pattern using nested for loops and a single break.

Instructions:

1. **Read** an integer n (use 3–20).
2. **Print** n rows.
3. **In each row** i, print characters from column j = 0 upward:
   - Print # normally.
   - When you reach the main diagonal (j == i), print \ **then break** the inner loop to end that row.

```
Enter size n (3..20): 12
\
#\
##\
###\
####\
#####\
######\
#######\
########\
#########\
##########\
###########\
```

Exercise 3

Use a for loop to sum integers, **skip** any number that contains the digit 5 (use continue), and **stop early** if the running sum **exceeds 1000** (use break).

Instructions:

1. **Read** an integer n (n ≥ 1).
2. **For** i = 1 to n:
   - If i contains the digit 5, **skip** it (continue).
   - Otherwise, add i to the running sum.
   - If the running sum **exceeds 1000**, **break** the loop.
3. **Print the final sum and the last value actually added** (call it last).



Thank You

Q & A